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Leif Weatherby: Two Autonomies of the Symbolic Order

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ÄNDERUNGEN

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2

Two Autonomies of the Symbolic Order: Data and Language in Neural Nets

Leif Weatherby

> Denn nicht *wir* wissen, es ist allererst ein gewisser *Zustand* unsrer, welcher weiß. [For it is not *we* who know, but above all a certain *state* in us, which knows.]
>
> —Kleist

The Data Hypothesis

To think with AI, our first order of business must be to jettison its guiding metaphor. No one knows what 'intelligence' is, and it functions as a regulative ideal for industry and research, not as a concept. The moment we let go of the term, we can refocus energy on the simulacrum we otherwise miss, the *stuff* that AI surfaces. Catharine Malabou argues that we have to abandon the old concept of human-individual 'intelligence' in order to confront the moving target of the 'negotiation' between 'the transcendental and the empirical' (Malabou 2019: 11). The 'mind' might still function as a 'shield' in the way that Sigmund Freud described (9), but we have sprung loose something else that negotiates on our behalf. The 'simulation of life' has blurred the borders 'between biological and symbolic life' (xvii). It is this region which, as we shall see below, language (and technology more generally) occupies anyway. (Malabou is borrowing Gilbert Simondon's framework, in which the creation of new *a prioris* is the vocation of technology.)[1] The 'symbolist' phase of AI failed to open up this border region, but the shift to data-hungry machine learning models has swung the pendulum heavily.

Neural nets, as Malabou recognises, have thrust a completely different philosophical genealogy into prominence. Where classical AI focused on the notion of 'physical symbol systems' and the logical manipulation of their symbols (Newell and Simon 1976), machine learning recalls the work of Jean Piaget,[2] the idea of the emergence of abstraction from perception in

hypothesis in this double sense undermines. 'Ground truth' is not 'fact' but a separate representation of fact in another semiotic system (usually language as expressive of concepts). The point is not that we cannot stabilise nets around accepted notions of truthful statements, but rather that when we do, we are black-boxing the dialogue between computation and language that underlies all nets (including in visual and other applications).

What I take to be the gold standard of interpretation of nets is Orit Halpern's notion of 'beautiful data', which shows how the dream of the net, long before its algorithmic implementation, already configured a world in which data streams are mutually interoperable and translatable, one that mimics the notion of the brain as such a transfer station, but then distributed into the world, forming an aesthetic condition (Halpern 2014). Many things follow from that condition, including Halpern's notion of 'derivativeness' in what she, Jeffrey Kirkwood, Patrick Jagoda and I have called 'surplus data' (Halpern et al. 2022), and what she further calls the 'smartness mandate' (Halpern and Mitchell 2022). Work on data is 'derivative' in the sense that a result obtained not from first-order empirical observation, but from a representation of those observations, is fed back into the world through logistical channels, rendering the very world we encounter 'derivative' of the data which was supposed to be a representation of that world. This feedback loop tends to render all judgement hypothetical. Rather than a world of stable fact about which hypotheses are made, the world of beautiful data looks more like a serial hypothesis from which the 'world' is first crystallised in its (usually overlooked) semiotic transformation with human systems of representation. The common channel of that semiosis is – and always will be – language.

Language as a Service, Attention as a Grid

A great deal of the confusion over the special class of nets called large language models (LLMs) boils down to the simple fact that language is the medium through which humans communicate with their computational machines. This has always been true. When Friedrich Kittler infamously called an essay 'There Is No Software', what he actually argued was that '[software] *would* not exist if computer systems did not – at least until now – need to coexist with an environment of everyday languages'. (Kittler 2014: 223). This phrasing makes it sound as though computing systems can take or leave language, but that is not the case. Language does not merely 'coexist' with computational systems – one need look no further than the input/output unit included in the 'First Draft of a Report on the Edvac' assembled by John von Neumann (generally taken to be the first explicit architecture for a stored-programme computer) to see that no digital system has ever been 'computational' in the absence of the linguistic environment (von Neumann 1993). It is all too easy to think of the edge of the diagram, which often reads 'I/O' for 'input/output', as not *really* part of the computer. But it is very hard to accept what would follow, namely

a 'computer' that does not render its computations at all, a true black box that could just as well be a slab of granite as a calculating engine. The relationship between computation and language is essential to every digital machine. Language and number may not get along in concepts or intuitions, they may even contradict one another in some essential way or ways. Kittler seems to suggest that they are forced into a relationship that obscures the computational core of the computer. But it is rather their internal relationship that allows for computing *in the first place*. They are not grafted onto each other using some third thing (the name of which Kittler leaves out anyway).[6] Just in the way that mathematics is the explanation of (among other things) numerical reasoning, computing is the rendering of linguistically meaningful results. Language is the medium of computation.

If it is hard to distinguish the simulation of induction and deduction in nets from their hypothetical core, it is even harder – and perhaps impossible – to separate their manipulation of language from simply language. This is because the role that labeling plays in classifying nets is taken over by language itself in LLMs. Linguistic applications in nets abandon the operation of supervision (the manual labeling of squirrels for the training dataset) to arrive at basic results, instead using the learning function to generate language, one word at a time. This inversion is crucial to thinking with AI, because it collapses the computation and its 'environment' – the results of the computation are language itself. This establishes a concrete version of a problem that I call 'the double autonomy of the symbolic order'. Where an image-classifying net produces a function that obviously is not the same as our conceptual grasp of the object at hand, a language-learning net learns to generate nothing other than language. Attempts to distinguish in some essential way between the 'real' language of humans and the merely 'apparent' language of nets make little sense on their face, and have disastrous consequences – as we shall see – for the analysis of these systems. Where a classifying net indicates, a language model generates icons using symbols. There is no such thing as hypothetical language, so nets learn language itself. They do so using the 'attention mechanism' at the base of the Transformer Architecture, the core of GPT systems.

Attention overcomes a problem that the net approach to AI had always had in linguistic applications, namely the problem of *memory*. Humans do not memorise language, they use it. I do not have a static store of all the sentences I've heard or read in my mind. I *use* language in the sense that I generate meaning from the combination of rules (grammar) and other meanings (words in grammatical context). The relationship between meaning and rules does not become explicit by that use. Meaning surfaces, but does not explain itself.

Early net-based approaches to language relied on memorisation (storage) over a sequence, leading to problems of misapplication, memory overload, and irrelevance. The most general format was the Recurrent Neural Net, in which the output of the net was fed back in as input, allowing the net to capture some of the structure in the linguistic strings. But this quickly maxed out

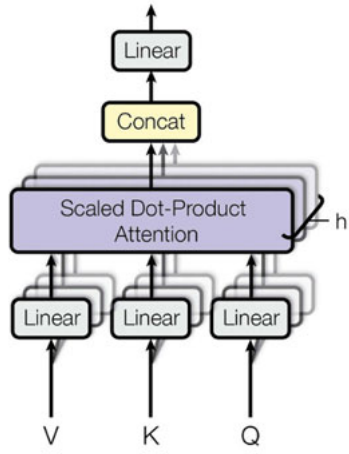


Figure 2. Attention head. Image from Vaswani et al. (2017).

the memory capacity of the grid, and led to spotty language use as the net 'forgot' pieces and rules of language that were distant from its current input. The creation of variably ranging memory in LSTM ('long short-term memory') solved part of this problem, but also hit barriers. The attention mechanism solves this problem by allowing the net to encode a one-to-all matrix for each word in the dataset. By including the positions of the words in the 'embedding' process (in which words are indexed to numbers), the problem of memory is solved by packaging the semantic relationships and their grammatical layout in a single 'pre-training'. Rather than jamming rules and words into some location, the net is allowed to 'attend' to real possibilities to move to from its current word.

One can think of this like a set of patterns pre-programmed into a grid of lights. Touch one light, and one shape turns on; another, and some other shape lights up. 'Attend' to one word and the grid's values shift to specify each other word's likelihood to be next. If the word I am attending to is 'see' and the word before it is 'I' then nouns of all kinds and the word 'that' will light up, meaning their value will range high. This set of higher values given a word in a sequence will be pooled by a 'temperature' feature – turned too high, this will produce too predictable a text, and too low, nonsense. But the Goldilocks temperature will produce […] the very 'everyday language' that the computation would otherwise have as its environment (see Wolfram 2023). And at this point, there is no 'classifying' function (which explains why LLMs are hit-and-miss for most serious reasoning tasks). The net produces, but does not tell us rules about, language. It uses language. The underlying mechanism is surely different in kind from the human use of language, but the property of generating meaning from the combination of rules and other meaning, is the same. There is no difference to speak of between the net's manipulation of data and the 'world' that data represents. The language that a model of

language produces is language. There can be no clear separation between hypothesis and deduction here, for if a net spits out a series of words that cohere as natural language, the difference between the semiotic function and the semantic function collapses – producing not just indications but instead meanings.[7] The language model distills language from its usual bundle with intelligence, affect, and other cognitive functions. The result of language is more language, and that is how language becomes a service.

I propose the term 'language as a service' to describe the world we already inhabit, in which language is on tap at the push of a button, a menu of options to activate across an infinite array of genres. LLMs loosen the relationship between intention, labour, and writing, leading media scholar Matthew Kirschenbaum to predict a 'textpocalypse' (Kirschenbaum 2023), as the textual rails of our civilization waver. We often hear now of the 'as-a-service-ification' of the economy, as seemingly everything comes unbundled as independent software packages. Nick Srnicek points, for example, to the platform economy model which Boeing uses to sell jet engines 'as a service', converting the commodity from engine to digital system. The engine itself is more or less rented, in this case, while what is actually sold is the proprietary software, including service and repairs. This both protects the code and the data of the selling company, while significantly altering the landscape of what counts as a commodity in heavy industry (Srnicek and De Sutter 2017).

Language as a service is at the extreme other end of the spectrum (as will be 'code as a service', which LLMs are already providing). For language to be provided pre-packaged as software, it must be produced outside of human minds. Where visual applications of nets externalise forms of classification and render images from labels (as DALL·E does), LLMs exploit the deeper relationship of word and number to generate the very stuff in which judgement must eventually be expressed and communicated: language. Perhaps nets will get better at exploiting the 'physical grammar' of movement in videos as time goes on, but for now the most important epistemological and economic result of these systems is the unrolling of massive amounts of linguistic redundancy as *more language*, surplus language. While the relationship between 'good' and 'safe' information and language is being worked out by the rentier class of the platform economy, however, we must urgently make two things clear. We must see that LLMs confirm and extend the idea that language is *already* artificial and independent of human minds, and for that reason we must move beyond single-dimensional analyses of AI, to what I call 'polythematic' semiotics.

The Double Autonomy of the Symbolic Order

If one had to point to a major achievement of the twentieth-century humanities, one could do worse than the concept of the 'autonomy of the symbolic order'. This idea can be traced back to the blurry transition between structuralism and post-structuralism. The idea that patterns of language, institution,

and psyche were independent of some 'ground truth' outside of the system in question animated a whole generation of French thinkers in the 'human sciences' (Dosse 1997).[8] I call this grouping 'the structuralism complex', since it forms a theory type distinguishable from others, family squabbles aside. Structuralism – which was intertwined at its height with cybernetics – arrived in the United States packaged in its rejection by a group of literary critics and philosophers whose influence is now embedded in virtually every branch of the humanities. One major concept that emerged from this moment was the idea that written language could not be accounted for by humanism. Terms and phrases like *différance* (Jacques Derrida), *différend* (Jean-François Lyotard), simulacrum (Baudrillard), and 'floating signifier' (Claude Lévi-Strauss) all share in the conviction that language is not restricted to thinking, cognition, or the understanding. The figure who most clearly and fully articulated this idea, however, was Jacques Lacan. Lacan divided Freud's psychological categories of the ego, the superego, and the id, into the 'registers' of the imaginary (where we make sense of our picture of reality), the symbolic (signs as opposed to images: numbers, letters, etc.), and the Real (which is completely inaccessible, defined as that which cannot be represented). As the cybernetics craze filtered through the personnel of structuralism into Paris, Lacan spent the year 1954-55 lecturing on the emancipation of the symbolic order in servomechanisms and early digital machines (Lacan and Lacan 1991). Patterns of symbols would, he said, emerge from machinic feedback (as cyberneticians often did, he used the example of automatic doors) in the physical world, but these would then redound into the structural patterns of human communication as inhuman entities. The symbolic order confronts us, from the psychoanalytic perspective, as alien from the jump. We are immersed in an order we did not generate, so that even if it is 'human', it does not sync with our expectations, desires, or sense of self. Writing and language are always alienating entities, and digital systems, Lacan predicted, would destabilise the structures of control we used to keep them 'human'.

The point that we need to bear in mind is that the structuralism complex (including poststructuralism) discovered this autonomy of the symbolic without reference to machines in the first instance. (This is how they differed from the manic energy that drove the dystopia and redemption narratives about machines in the Weimar era.) Structuralism was driven by the discovery of *really existing concrete patterns of signification that could not be reduced to some first cause*. The problem then quickly became how to think about technology's impact on that already-autonomous artificial order of signs. Lacan captured this double autonomy in the slogan 'the world of the symbolic is the world of the machine' (Kittler, Johnston, and Johnston 1997). This problem is one of the most pressing in the era of AI.

This line of thought was one major strand in the founding of the modern discipline of media theory. Friedrich Kittler laid the groundwork by breaking with the thesis that media are 'extensions' of human capacities, most famously

articulated by Marshall McLuhan. The extension thesis, by keeping media in constant relation to human capacities, disallows the autonomy of media effects. Kittler went to great rhetorical and historiographical lengths to argue that this was a false starting point for the reproduction of sound, image, and 'writing'. Writing, as an activity done by hand, mind, and emotion working together, gained a storage and reproduction medium external to the writing itself in the printing press. During this period (what McLuhan called the 'Gutenberg Galaxy'), writing remained, according to Kittler, the only way to store and reproduce images and sounds (by means of the imagination) (Kittler 1990). With the typewriter, the point of production was altered, because the set of symbols was no longer held in and between minds, but laid out in the keyboard. The printing press was miniaturised at the labour interface of writing. The 'symbolic order' could be seen as a natural part of the human mind – or as the non-natural essence of humanity's non-fit with nature – only until the relative autonomy of language was made material. At that point, language had to be seen as its own instance, bearing its own force – this is what I understand Kittler to mean when he says that Lacan brings psychoanalysis up to date with the media (Kittler, Johnston, and Johnston 1997).

This autonomy, however, is not restricted to letters, and when it took its incipient computational form, the question concerning the symbolic order became more urgent. Erich Hörl's elegant account of how the formalisation of thought led to an 'ecology' of such independent symbols follows this Lacanian trajectory (Hörl, Nancy, and Schott 2018), as does Bernhard Siegert's notion that the 'passage of the digital' is the intervention of the symbolic in the Real, where it has been set loose without possibility of recuperation into our understanding, creating a 'schizoid' binary (Siegert 2003). Siegert calls this type of operation *Kulturtechnik* or 'cultural technics',[9] a technique that cultivates a difference (Siegert and Winthrop-Young 2015). The value of this move is clear: differences of this kind are not introduced into some blank-slate state of nature in which all is pre-stabilised. But there is a danger here, too, which is that we trivialise the way in which one difference is built on top of another difference in sequence. The 'autonomy of the symbolic order', as I am deriving it here, stands for a meta-difference in which an order that is already neither natural nor fully 'human' (in the sense of 'held within human understanding') is displaced *again*. To deal with that situation, you need not just 'cultural technics', but instead what Jeffrey Kirkwood calls 'semiotechnics', 'a theory of technical systems focusing on their management of absences, breaks, and gaps for the production of meaning' (Kirkwood 2022, 9). The reason this revision is crucial is that the relationship between the machine and the order of signs reveals an integration of two orders of difference with each other. It is not in the gap between representation and thing that we will find answers, but in the merging and contradictions between two orders. This is a problem of form, syntax, or reason, but only if each of these terms is stripped of it's a priori, Gods-eye status – the real relation between shapes and complexes of signs and reasons

is the common ground of semiotics and dialectics. When we arrive at this type of complexity, we need to find a descriptive language that lives up to it – but the tendency is, instead, to revert to a panicked humanism that is little more than a remainder.

Remainder Humanism

It was the literary theorists who rescinded the genteel gesture of the Turing test for actual human authors. Turing's famous test was based in social recognition: if I cannot tell if I am talking to a human or a machine, then a machine in that position qualifies as intelligent. The repeated announcements that the author was 'dead' refuse to animate text by means of that recognition, scrutinising it without the imaginary interlocutor. But these announcements seem today to have been half-hearted at best. It is as though the 'autonomy of the symbolic order' never *landed* across a wide swath of disciplines. It hovers, we learn it, it sometimes makes partial contact with one or another object. But its Age of Aquarius has dawned with GPT systems, and we see a great recoiling into the remaining corners of viable humanism. I call this 'remainder humanism', a failure to take seriously the cultural-systematic state of play. Every appeal to the 'flattening' of 'complex' and 'human' and 'embodied' secrets allegedly absent from digital systems fails, *not* because those things do not exist, and not *even* because they are not manipulated, harmed, and even 'murdered' (in Jonathan Beller's hyperbole (Beller 2017)) by those systems, but because those terms do not show us why those harms are possible, and what the laws of their unfolding will be. No binary system will describe the interaction of two orders of difference. When Franco 'Bifo' Berardi writes that 'digital networks […] have penetrated the social organism […] but the two levels cannot synchronize', we are witnessing little more than resistance to AI (Berardi 2023). This remainder humanism is the theoretical equivalent of painting oneself into a corner. No amount of insistence that digital and human systems 'fit badly' together will change the concrete situation, in which the merger of those systems is the most glaring fact. This goes for their 'alignment' too, which occurs in fact but not as we would explicitly want it to, since that desire is also expressed in utterly naïve binary terms.

Noam Chomsky has argued that GPT systems have no substantial relationship to language whatsoever. In a high-profile op-ed, Chomsky conflates the question of intelligence with that of language, attempting to show how humans use language cognitively, seeking and establishing truth and reasoning morally, in ways that algorithms can't (Chomsky 2023). This argument is both old (Chomsky has been making a version of it since 1957) and strangely off-target. No one thinks – at least, no one should think – that GPT systems do language the way humans do language. And in restricting language to the synthetic functions of the human brain, Chomsky epitomises the humanistic recoil against the autonomy of the symbolic. We do not need to disagree with

his account of the human use of language to disagree that this use of language is language as such. The sheer bravura of dismissing more than a trillion words of human-produced language on the grounds that it is then reproduced algorithmically is the perfect fit for the arrogance of the AI movement itself, which takes no account of the harm, but also none of the interesting non-'intelligence' effects, of its systems. But even this grandiose claim is a remainder – a remainder of a separation between cognition and language that is only obscured by the systems that now produce the latter without the former.

Chomsky seems to confuse the generation with the use of language. Humans do both, of course, but when we cognitively synthesise words in order to arrive at an idea we hold to be true or good, we are mostly using language. It is not at all clear that my typing this sentence 'produces' language in a strong sense. It contributes to the overall tendency of English, I suppose, but as I choose each word I do so with genuine semantic limitations. I can replace 'produces' in the previous sentences with 'generates', but not with 'garbles' or 'vgfrdb'. To 'produce' language in the sense implied by Chomsky is somehow to generate it *as if for the first time*, in keeping with the alleged device in the brain that first gave rise to language. Whether I am following external or internal rules remains undecided, mixed – for we are not at the origin of language. An interface like ChatGPT might be *even further* from that origin, and in spite of – even because of – this, reveal even more about what language is than our cognitive systems can. The act of speaking is social and technological, not human in some sense innocent of either.

The humanistic view of language denies not the double autonomy of the symbolic order, but the very social-artificial aspect of language as such. Meaning is not made in the mind; at a minimum it must be between minds. But this already provides the wedge that allows the strange admixture of technology and language.

Kant, Kittler… and Kissinger?

In what is perhaps to date the greatest exemplification of the meme 'the worst person you know just made a great point', Henry Kissinger has adopted the opposite tack from Chomsky's (Kissinger et al. 2023). Kissinger thinks that AI is a serious global problem. This is because nets have broken out of the mould of previous ancillary digital systems, and in doing so, have plunged us into the desert of the Real.

This is not an exaggeration. Kissinger thinks that neural nets are scouring areas of existence that human minds cannot, in principle, have access to. For that reason, the change that they represent is total, a change in the very idea of epistemology as much as an attendant shift in global power relations. He writes with his co-authors that AI is overcoming the self-understood limitations that the Enlightenment put on reason. AI is reaching past Kant's 'essential distinction between the thing in itself and the unavoidably filtered world

we experience', and beginning to provide an alternative means of accessing – and thus understanding – reality' (Kissinger et al. 2021: 43). Here, Kissinger is unwittingly Kittlerian. Computation reaches into the depths of the Real, but rather than returning with some piece of it packaged for human understanding, shifts the very structure of our felt reality in ways that reverberate beyond our understanding. The future, Kissinger writes, will be 'a collaboration […] with a different kind of technical entity but with a different kind of reasoning' that will 'precipitate a transformation in metacognition and hermeneutics – the understanding of understanding – and in human perceptions of our role and function' (Kissinger et al. 2023). In fact, Kissinger here goes one step beyond Kittler and his school, with their famous rejection of hermeneutics. He suggests that a 'dialectic' (his word) between human understanding and neural nets will be established, one that we must urgently begin to guide even as it emerges.

Here we have a most wonderful formula: Kant, Kittler, and Kissinger. We might capture their collective rejection of 'remainder humanism' by supplying them with a common response to the X-Files' slogan 'the truth is out there', to which they say: no, it's not. The truth is in negotiation with this other form of mediated knowledge. The virtue of this motley crew of Kant, Kittler, and Kissinger is that they are able to jettison the metaphysics of intelligence entirely. Downstream from the loss of this ballast is a redefinition of language that recognises the importance of what Chomsky is saying – it is our hook for meaning, for generating semantics and for doing hermeneutics – but without the reification of a limited notion of thinking (essentially an individualistic one) as the essence of that language.

But even if the truth is mediated for him, you can feel the outmoded policy-control ambition in Kissinger. He has an unusually clear vantage-point, for obvious reasons, on the global logistical system made of net-based AI, and sees that this will not fit with current statecraft or capitalism.[10] But just in proportion with this clarity is Kissinger's inability to think beyond the mid-century problem of policy and control. This is its own form of remainder humanism, in the realm of practical rather than pure reason. But doesn't this dream of control, in which policy magically gets out ahead of our ultra-complex systems, sees them for what they are, and guides them into 'alignment' with a fantasised consensus, apply to most of our critical apparatuses too? Precisely because the centrist state has adopted data policy as the furthest horizon of its political imagination, we see many attempts to regulate the problems out of existence funded at the highest levels. But think-tankisation will not solve the metaphysical problems that critical theory was engineered to address. The truth is not 'out there', but just because it is caught in systems of signification does not make it easy to control. That fact is so obvious that it feels strange to have to point it out, yet we seem unable to focus our attention on the theoretical half of the problem long enough to address it. That, I want to argue, is because of what I call 'the monothematic sublime'.